

Galaxy morphology in CANDELS: addressing evolutionary changes across $0.2 \leq z \leq 2.4$ with hybrid classification approach

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Accepted 2025 April 11. Received 2025 March 19; in original form 2024 October 12

ABSTRACT

Morphological classification of galaxies becomes increasingly challenging with redshift. We apply a hybrid unsupervised-supervised method to classify $\sim 14\,000$ galaxies in the Cosmic Assembly Near-IR Deep Extragalactic Legacy Survey fields at $0.2 \leq z \leq 2.4$ into spheroid, disc, and irregular systems. Unlike previous works, our method is applied to redshift bins of width 0.2. Comparison between general versus bin-specific models applied to a wide redshift range and reveals significant differences in galaxy morphology beyond $z > 1$ and a consistent ~ 25 per cent disagreement. This suggests that using a single model across wide redshift ranges may introduce biases due to the large time intervals involved compared to galaxy evolution time-scales. Using the FERENGI code to assess the impact of cosmological effects, we find that flux dimming and smaller angular scales may lead to the misclassification of up to 18 per cent of disc galaxies as spheroids or irregulars. Contrary to previous studies, we find an almost constant fraction of discs (~ 60 per cent) and spheroids (~ 30 per cent) across redshifts. We attribute discrepancies with earlier works, which suggested a decreasing fraction of discs beyond $z = 1$, to the biases introduced by visual classification. Our claim is further strengthened by the striking agreement to the results reported by Lee et al. using an objective, unsupervised method applied to *JWST* data. Exploring mass dependence, we observe an ~ 40 per cent increase in the fraction of massive ($M_{\text{stellar}} \geq 10^{10.5} M_{\odot}$) spheroids with decreasing redshift, well balanced with a decrease of ~ 20 per cent in the fraction of $M_{\text{stellar}} \geq 10^{10.5}$ discs, suggesting that merging massive disc galaxies may form spheroidal systems.

Key words: methods: observational – galaxies: evolution – galaxies: photometry – galaxies: structure.

1 INTRODUCTION

Morphology has been a fundamental probe of galaxy formation and evolution. Morphological features reveal dynamical mechanisms shaping galaxies (Peebles 1969; Teklu et al. 2015). For example, high angular momentum in a protogalactic cloud leads to the formation of a rotating disc structure, resulting in spiral galaxies with well-defined discs and spiral arms. Conversely, low angular momentum causes a more isotropic collapse, forming spheroidal structures typical of elliptical galaxies. Irregular distributions of angular momentum can give rise to morphologies in between spheroidal and disc galaxies, often referred to as irregular systems.

Morphology is also connected to variations in star formation efficiency. In disc galaxies, differential rotation promotes star formation, particularly in spiral arms where density waves compress the gas (Cimatti, Fraternali & Nipoti 2019). In spheroidal galaxies, rapid gas collapse can lead to intense, but short-lived star formation, resulting in older stellar populations in present-day observations and negligible ongoing star formation (Strateva et al. 2001; Wetzel, Tinker & Conroy 2012; Trussler et al. 2020).

Furthermore, the morphology–density and morphology–cluster-centric distance relations indicate a higher proportion of early-type galaxies (ellipticals and S0 galaxies) in denser regions of galaxy clusters and closer to cluster centres compared to low-density fields (Dressler 1980; Dressler et al. 1997), indicating that environment can also leave significant imprints in galaxy morphology. In the local Universe, data from the Sloan Digital Sky Survey (SDSS; York et al. 2000) have shown that elliptical galaxies are predominantly found at high stellar masses and occupy specifically the red sequence region in the star formation main sequence diagram, whereas spiral and S0 galaxies are mainly found in the blue cloud and green valley, respectively (Schawinski et al. 2014; Sampaio et al. 2023).

At higher redshifts, the *Hubble Space Telescope* (*HST*) is a crucial tool for characterizing galaxy morphology due to its depth (~ 27 for the *F814W* filter), high spatial resolution, and broad wavelength coverage from optical to near-infrared. Studies using *HST* data have consistently reported that galaxies at $z \sim 2$ often exhibit disturbed or irregular features, suggesting a predominance of irregular galaxies at early stages of the Universe (Conselice 2014). However, reliably characterizing galaxy morphology at high redshifts is challenging due to cosmological effects and observational limitations. For instance, decreasing angular size and flux dimming with redshift. Different methods yield significantly different results. For instance, Bruce et al. (2012) characterize the Sérsic index of massive galaxies

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[$\log(M_{\text{stellar}}/M_{\odot}) \geq 11$] and find a significant increase in the fraction of disc-like galaxies from 15 per cent at $z = 0$ to 80 per cent at $z = 2$, while the fraction of spheroidal galaxies decreases from 85 per cent to 20 per cent over the same range. In contrast, Mortlock et al. (2013) visually classify galaxies with $\log(M_{\text{stellar}}/M_{\odot}) \geq 10$ at $z > 2$ from the UltraDeep Survey (UDS) and find a higher fraction of spheroidal galaxies (40 per cent) compared to disc-like systems (10 per cent). These discrepancies highlight the impact of different classification methods on morphological analyses at high redshift.

Parametric approaches, such as light profile fitting, depend on well-behaved light distributions, which are often not the case at high redshifts. For instance, fitting a Sérsic law to an irregular galaxy light profile is senseless, as it has subtleties not taken into account in the adopted law. On the other hand, non-parametric approaches, such as the Concentration + Asymmetry + Smoothness (CAS) system, do not rely on any assumption about the galaxy light profile, thus being able to characterize even irregular galaxies (Barchi et al. 2020). However, it has been shown that the CAS metrics have limitations in distinguishing between disc- and bulge-dominated galaxies. As an alternative, Kolesnikov et al. (2024, K24 hereon) show that a system combining Entropy + Gini index + Gradient pattern asymmetry (EGG) is more reliable in separating these two morphological types. Nevertheless, whichever approach is used, pre-processing (defining the cut-out, background subtraction, cleaning of secondary objects, and creating a segmentation mask) has always an effect on either the fitting of the light profile or the measurement of the metrics and, as galaxy angular scale decreases with redshift, resulting in a smaller number of pixels for a given segmentation mask.

Alongside parametric and non-parametric methods, machine learning (ML) and deep learning (DL) algorithms are gaining more attention each year. The benefits of these tools, such as scalability, speed, and accuracy, provide substantial improvements in classification and analysis. Multiple papers use ML/DL algorithms to achieve automated classifications (Ferrari, de Carvalho & Trevisan 2015; Khalifa et al. 2017; Domínguez Sánchez et al. 2018; Primack et al. 2018; Khan et al. 2019; Barchi et al. 2020; Tohill et al. 2021; Walmsley et al. 2022, 2023; Cheng et al. 2023). More recently, works (Wei et al. 2023; Popp et al. 2024) apply transfer learning and backbone augmentation with attention and deformable convolution. Such methods offer numerous benefits (Dai et al. 2017; Bello et al. 2019; Zhao et al. 2024). However, sophisticated methodologies will only be effective if the ground truth catalogues are well defined, since, being a supervised approach, the model will inherit all imperfections of the training set.

Currently, one way to define ground truth data sets is through citizen science projects such as Galaxy Zoo (Lintott et al. 2011) or Zooniverse (Simpson, Page & De Roure 2014). In addition, small teams of astronomers can classify smaller amounts of data with specific goals in mind (Cassata et al. 2005; Delgado-Serrano et al. 2010; Mortlock et al. 2013; Talia et al. 2014; Kartaltepe et al. 2015). In both scenarios, the classification is done by visual inspection. While the human eye possesses a great capacity for integration and feature extraction, it is also susceptible to bias and subjectivity, where small, faint, or distant galaxies could be classified differently by different individuals. In this paper, we argue that visual bias has a serious impact on morphological analysis due to its propagation to ML models, especially at high redshifts.

Combining these two critical points, bias and time demands, we note a gap that needs to be filled by a method that will be fast, general, objective, reproducible, and applicable to different data sets. In this work, we introduce a novel method to classify galaxy morphology at $0.2 \leq z \leq 2.4$ using a hybrid unsupervised-supervised approach.

We apply this method to galaxies from the Cosmic Assembly Near-IR Deep Extragalactic Legacy Survey (CANDELS) observed in the *F814W* filter by the *HST* Wide Field Camera (WFC). We implement updates in our method in comparison to K24, in order to provide reliable classification for high-redshift galaxies. We also examine the biases introduced by using a single model across a wide redshift range and investigate the fraction of disc and spheroidal galaxies as a function of redshift, comparing our findings with previous studies.

This paper is organized as follows: in Section 2, we present the sample selection and the catalogues used to retrieve galaxy properties; in Section 3, we provide a brief description of the hybrid unsupervised-supervised method and present some modifications with respect to the methodology employed in K24; in Section 4, we present a detailed investigation on the performance of convolutional neural network (CNN) classification due to redshift-related effects; in Section 5, we present our results and discuss observed trends in a cosmological context; and Section 6 summarizes the main results of this work. Throughout this paper, we adopt a flat Lambda cold dark matter cosmology with $[\Omega_M, \Omega_\Lambda, H_0] = [0.31, 0.69, 68 \text{ km s}^{-1} \text{ Mpc}^{-1}]$ to be consistent with *Planck 2018 cosmology* (Planck Collaboration VI 2020) and report the magnitudes in the AB system. AB is a photometric system designed to provide uniform magnitude across all wavelengths, such that a source with a constant flux per unit frequency of 3631 Janskys has an AB magnitude of 0 in all filters.

2 SAMPLE SELECTION AND DATA USED

In this paper, we utilize a combination of fields from CANDELS (Faber 2011; Grogin et al. 2011; Koekemoer et al. 2011) as our primary data sample. CANDELS is a 902-orbit legacy programme designed to study galaxy formation and evolution over a wide redshift range using the near-infrared WFC3 camera on the *HST*, which obtains deep imaging of faint and distant objects. To date, CANDELS has imaged over 250 000 distant galaxies within 5 strategic regions: The Great Observatories Origins Deep Survey South and North (GOODS-S and GOODS-N), UDS, Extended Groth Strip (EGS), and Cosmic Evolution Survey (COSMOS), covering a combined area of approximately 0.22 deg^2 (e.g. Barro et al. 2019).

We restrict our sample to the COSMOS, EGS, UDS, and GOODS-S fields, due to the uniformity in photometric catalogue and methodology to estimate stellar masses (Santini et al. 2015; Nayyeri et al. 2017; Stefanon et al. 2017). In a few words, stellar mass estimates are derived by performing SED fitting on the observed multiwavelength photometry (ranging from infrared to ultraviolet) with stellar population synthesis templates (Fioc & Rocca-Volmerange 1997; Bruzual & Charlot 2003; Maraston 2005; Bruzual 2007). To avoid biases due to the choice of minimization method and/or adopted fitting code, the effort of 10 different teams is combined, each one adopting their preferred fitting code, assumptions, priors, and parameter grid. Therefore, the final results using the same underlying stellar isochrones reduce the systematic uncertainties associated with the fitting code and stellar population choice. Hereafter, we use the median stellar mass for the galaxies in our sample. Although the original CANDELS photometric catalogue classifies galaxies using the *F160W* filter ($\lambda_{\text{eff}}^{F160W} = 14445 \text{ \AA}$), we select only galaxies that were observed using the *F814W* filter ($\lambda_{\text{eff}}^{F814W} = 8050 \text{ \AA}$) from the Advanced Camera for Surveys (ACS). This follows from a better resolution available for *F814W* observations, namely a pixel scale of $0.05 \text{ arcsec px}^{-1}$, in comparison to $0.1 \text{ arcsec px}^{-1}$ for the WFC3.

In Figs 1(a) and (b), we present stellar mass and apparent magnitude distributions, respectively, for each field (different colour) and when combined (in dark). Notably, the distributions are similar

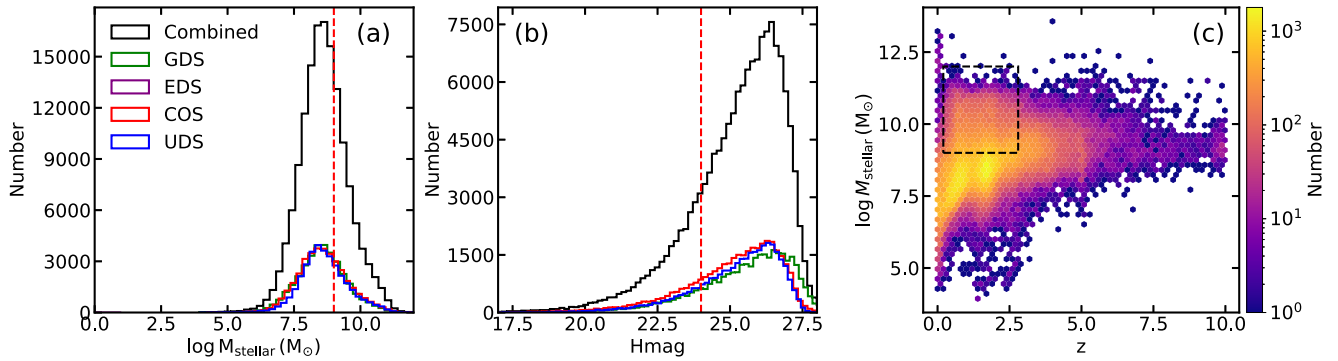


Figure 1. Panel (a) displays the stellar mass distribution across each field, whereas panel (b) presents the distribution of apparent magnitudes in the H band. Dashed horizontal lines demonstrate cuts we applied to select data for our sample. Panel (c) shows the stellar mass distribution of galaxies in the CANDELS fields, with the selection criteria indicated by the dashed black rectangle.

for all the four fields; thus, we apply the same thresholds irrespective of field. First, we select only galaxies with stellar masses greater than $10^9 M_\odot$ to avoid including dwarf galaxies in our sample, since these kind of objects have a higher uncertainty associated with SED (Spectral Energy Distribution) fitting (Santini et al. 2015). Secondly, we limit our sample to galaxies brighter than $H = 24$. We highlight that, despite utilizing ACS $F814W$ images, we apply magnitude thresholds based on the $F160W$ filter, which was originally used to detect and characterize galaxies in the four fields (Barro et al. 2019). The value $H = 24$ is a conservative threshold that follows from works suggesting that both non-parametric morphological indices and automated algorithms struggle to reliably characterize galaxy morphology for fainter systems (Grogin et al. 2011; Kartaltepe et al. 2015). In Figs 1(a) and (b), the red vertical line shows the adopted thresholds. In Fig. 1(c), we show the number of objects as a function of stellar mass and redshift for the four fields combined. Notably, there is a significant decrease in the number of objects more massive than $10^{10} M_\odot$ above $z \sim 2.4$. Therefore, we limit our analysis to galaxies in $0.2 \leq z \leq 2.4$. Our combined selection criteria are illustrated by the dashed black rectangle, which comprises 16 718 galaxies available in $F814W$ ACS/WFC.

2.1 Data preparation strategy

As outlined in K24, data preparation is a fundamental step, especially when estimating the non-parametric indices. Using the combined drizzled $F814W$ fields provided by the CANDELS survey,¹ we create cut-outs centred in each galaxy of the sample and with size empirically defined as $K \times \text{FLUX_RADIUS}$ (in pixels), where FLUX_RADIUS is the radius containing 50 per cent of the galaxy total light in the $F814W$ band. We use $K = 5$ for cut-outs used for metric extraction and $K = 3$ for classification using DL. The difference is justified by the need of background subtraction during metric extraction, which requires a wide ‘empty’ area around the galaxy. On the other hand, for the DL classification, only the galaxy image is needed. We remove 1982 galaxies from our sample, as they are located close to the CANDELS follow-up edges or suffer from the presence of nearby bright objects, which may cause biases in the non-parametric metric’s estimation. Even though we remove 11 per cent of the total number of galaxies, we ensured that this does not alter the original stellar mass and $H\text{mag}$ distribution. After

this exclusion, our sample consists of 14 736 galaxies. We further separate galaxies into redshift bins varying from 0.2 to 2.4 in steps of 0.2, where the step is selected to guarantee at least 500 galaxies in each bin. We show the number of galaxies in each redshift bin in Fig. 2. This separation in redshift bins is particularly relevant to our method, which is described in detail in Section 3.

A word of caution is needed regarding the redshift bin width choice. We consider three main factors to define the redshift bin size. First, the redshift bin should contain enough galaxies (≥ 500) for the unsupervised method to work properly. Secondly, we consider the associated error of photometric redshift, especially above $z \geq 1.5$, where the uncertainty is ~ 0.2 , therefore preventing a smaller bin size. Finally, the choice of redshift binning is based on the idea that a wide redshift range may cover galaxies at different morphological evolutionary stages, such that different galaxy evolution-related processes are blended in a coarse model. Quantitatively, we are covering roughly 11 Gyr of cosmic history from $z \sim 0.2$ to ~ 2.4 . The selection of 0.2 as a redshift bin ensures an average sampling of ~ 1 Gyr per bin. We further investigate how our results depend on redshift bin size choice, by repeating the method described in Section 3 using a redshift bin width of 0.4. The fraction of spheroidal, discs, and irregulars change by, on average, 6, 6, and 1 per cent, respectively. The variations are considerably smaller than the uncertainties in the morphology fraction itself. Therefore, we claim that our results are roughly invariant due to the redshift bin width choice, as long as it is consistent with the time-scales related to galaxy evolution-related processes.

Using the original cut-out for each galaxy, we perform background subtraction, cleaning, and segmentation mask. The last is particularly relevant as it defines the pixels belonging to the galaxy, thus the ones that are used in the measurement of non-parametric indices. Since our sample covers a wide redshift range, we define our segmented mask as the best-fitting ellipse with semimajor axis equal to the Petrosian radius, which is irrespective of redshift.

2.2 The morphological system $M_{20} + \text{EGG}$ in a nutshell

In this work, we use the EGG system as a first approach to characterize galaxy morphology. This choice follows from a better performance of this system in comparison to the CAS system commonly adopted in the literature (e.g. K24). In addition, we include the second moment of light distribution M_{20} , defining the MEGG system. The inclusion of M_{20} follows from promising results with SDSS and HST morphological separations in Lotz, Primack & Madau (2004) and

¹All CANDELS fields can be accessed at <https://archive.stsci.edu/hlsp/candels>.

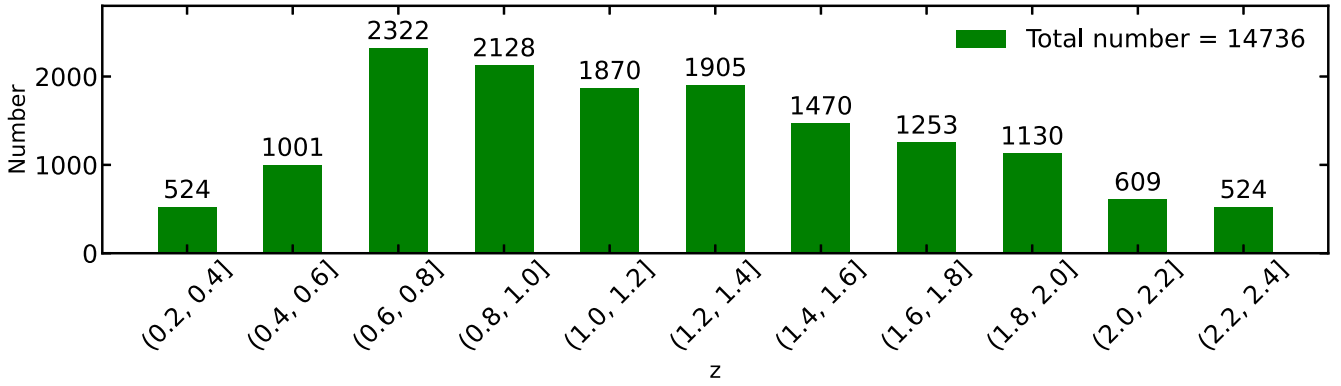


Figure 2. Final galaxy selection for hybrid classification after applying the cuts of $H_{\text{mag}} = 24$ and stellar mass greater than $10^9 M_{\odot}$, followed by cleaning, segmentation of cut-outs, and metric measurement. The data are divided into redshift bins, with each bin containing galaxies within a specified redshift range, treated as an independent data set.

Cheng et al. (2021). Moreover, the additional metric results in a more refined set of non-parametric indices that may help to define better and unbiased training samples (see Section 3). Below, we briefly describe the metrics composing the MEGG system, emphasizing that we estimate the metrics for each galaxy in the best-fitting ellipse with semimajor axis equal to the Petrosian radius, in order to guarantee robustness irrespective of redshift.

(i) **Entropy (E):** The entropy of information characterizes the distribution of pixel values in an image. Originating from Shannon entropy, it captures the randomness or unpredictability inherent in the images' information content (Bishop 2007; Ferrari et al. 2015). Following this definition, spheroidal galaxies have lower entropy in comparison to disc systems.

(ii) **Gini coefficient (G):** Originally used in economics to represent wealth distribution, the G coefficient has been adapted for galaxy morphology to measure the relative flux distribution across pixels corresponding to a galaxy. While it correlates with concentration, it does not necessarily presume the brightest pixels to be centrally positioned in the galaxy image (Abraham et al. 2003). Therefore, a high Gini index indicates a large concentration of light in a small number of pixels, which is characteristic of spheroidal galaxies. Conversely, a low Gini suggests a more even distribution of light across the galaxy, which is mainly seen in disc systems.

(iii) **Second gradient moment (G_2):** Gradient pattern analysis is a method proposed to gauge gradient bilateral asymmetries in a numerical grid. Within galaxy morphometry, the second gradient moment, represented as G_2 , stands out. It proves especially valuable, showcasing its potential to differentiate spheroids from disc systems more effectively than traditional morphometric metrics. For instance, G_2 achieves about 90 per cent accuracy in separating galaxies using data from the SDSS-Data Release 7 catalogue (Rosa et al. 2018). The computation of G_2 starts by calculating the gradient field in the X and Y directions of the image. With these two components at each location, we attribute a vector (characterized by module and phase) for each pixel. Then, we compare the magnitude and direction of equidistant vectors from the galaxy centre. Pairs that are symmetric (same magnitude but opposite direction) are excluded, resulting in a field with only asymmetric vectors. The next step involves calculating the number of and the sum of the asymmetric vectors (see K24 for more details on how G_2 is estimated).

(iv) **The second moment of light (M_{20}):** The M_{20} parameter is calculated from the second-order moment of the brightest regions of

a galaxy. The total second-order moment, M_{tot} , of a galaxy's light distribution is defined as

$$M_{\text{tot}} = \sum_i M_i = \sum_i f_i [(x_i - x_c)^2 + (y_i - y_c)^2], \quad (1)$$

where f_i is the flux in the i th pixel, (x_i, y_i) are the coordinates of the i th pixel, and (x_c, y_c) are the coordinates of the galaxy's centre. The M_{20} parameter is then defined as the normalized second-order moment of the brightest 20 per cent of the galaxy's flux,

$$M_{20} = \log \left(\frac{\sum_i M_i}{M_{\text{tot}}} \right), \text{ while } \sum f_i \leq 0.2 F_{\text{tot}}, \quad (2)$$

where F_{tot} is the total flux of the galaxy, and f_i is the flux in a given pixel. To calculate M_{20} , we (1) rank all pixels by their flux values in descending order; (2) calculate the cumulative flux F_{cum} until it reaches 20 per cent of the total flux F_{tot} ; (3) sum the second-order moment M_i for all pixels contributing to this cumulative flux; and (4) normalize this sum by the total second-order moment M_{tot} . With this definition, lower values of M_{20} indicate that the brightest regions are more centrally concentrated – related to spheroidal galaxies, while higher values suggest a more extended flux distribution – suggesting disc galaxies.

3 AN UPDATED VERSION OF THE HYBRID APPROACH

In this section, we present the method and processing pipeline, along with its updates and refinements in comparison to K24. It is important to note that extracting morphological information for the galaxies in this high- z sample is significantly more challenging due to the degradation effect (which is explored in Section 4). For this reason, we revise some steps in our method.

3.1 SOMBRERO and unsupervised labelling

We use the non-parametric indices' vector ($\mathbf{v} = [M_{20}, E, \text{Gini}, G_2]$) as input for a Self-Organizing Maps (SOM) clustering algorithm, implemented through the SOMBRERO R package (Villa-Vialaneix 2017). We choose SOMBRERO primarily due to its ability to process numeric data sets, given that we supply the metrics as numerical quantifiers of galaxies' physical properties. Moreover, the SOMBRERO package offers a robust functional foundation for working with data flow, distinguishing it from other implementations and clustering methods. As pointed out in K24, without meticulous cluster selection,

SOMBRERO demonstrates classification performance comparable to other clustering algorithms.

Initially, all galaxies in our data set are unlabelled, and the description of morphology comes from the metrics. SOMBRERO then defines a set of N_{clusters} , each initialized with a random value for each metric. For each galaxy, SOMBRERO calculates the Euclidean distance between the metrics vector $\mathbf{v}$ and the i th cluster weight vector $\mathbf{w}_i$. The cluster with the smallest distance is identified as the best matching unit (BMU), and the weights are adjusted to become closer to the galaxy's metrics, following the equation

$$\mathbf{w}(t+1) = \mathbf{w}(t) + \theta(t, r) \cdot \alpha(t) \cdot (\mathbf{v} - \mathbf{w}(t)), \quad (3)$$

where $\mathbf{w}_i(t)$ is the weight vector of the cluster at step t , $\mathbf{w}(t+1)$ is the updated weight vector, $\mathbf{v}$ is the metrics vector of the current galaxy, $\alpha(t)$ is the learning rate – i.e. a quantity that tells us how much the weights change in response to each galaxy, and typically decreases over time, $\theta(t, r)$ is the neighbourhood function (often a Gaussian) – which defines how the update of weights decreases for nodes farther from the BMU, and decreases with BMU distance and time, and r is the radius of the neighbourhood around the BMU affected by the update.

In early stages, weights [higher $\alpha(t)$ and larger neighbourhood radius r] suffer larger changes, allowing the SOM to organize itself according to the data's structure. As the training progresses, $\alpha(t)$ decreases and r may also decrease, leading to finer adjustments and stabilization of the map. In our pipeline, we use the Letremy implementation of distance and radius calculations. The Letremy distance is a type of measure designed to handle data by measuring the dissimilarity between two variables. It takes into account the co-occurrence frequencies of categories across the data set.

More details can be found in the original publications (Cottrell, Ibbou & Letrémy 2004; Cottrell & Letrémy 2005).

The final result of the SOM algorithm is a grid of clusters, each containing galaxies with similar metric distributions. The clusters that demonstrate the most defined metric distribution generally contain the most distinct objects (see fig. 12 of K24). We term these clusters as ‘prominent clusters’ and base our label assignment on them.

3.2 Prominent cluster selection

In comparison to K24, the prominent cluster selection process has undergone slight changes to enhance robustness and flexibility. Instead of using the median with a shift to select prominent clusters based on each metric, we implemented the IsoData (Iterative Self-Organizing Data Analysis Technique Algorithm) clustering algorithm (Velasco 1980; Lloyd 1982; Jensen 1996) that finds the threshold for each metric. The IsoData algorithm is particularly useful for scenarios where the number of clusters and their initial centroids are not known a priori, allowing the algorithm to dynamically adjust these parameters during its execution. In our case, we use the IsoData algorithm to analyse and partition the data based on a specified metric within a data frame. This partition signifies the distinction between the two morphological classes, determining at which value a given metric indicates whether a galaxy is classified as a disc or spheroidal.

Notably, clustering algorithms such as SOMBRERO can be affected by pre-existing bias in the data set. In K24, we provide a detailed discussion on mitigating biases, and in this work we have implemented the following additional procedures:

(i) Both E and G_2 depend on tunable parameters. The former depends on the number of bins used to build the histogram of counts in a galaxy image, while the second relies on the module and phase tolerance when comparing diametrically opposite vectors. Thus, we tune these parameters in each redshift bin, individually. This adaptive tuning allows the capture of variations in the data set across the whole redshift range, ensuring that the clustering reflects genuine structural differences rather than artefacts from a coarse approach.

(ii) To reduce the sensitivity to the choice of grid size and metric weighting, we run the SOMBRERO algorithm varying the grid size by ± 2 units from the ideal value L_{grid} , defined by $L_{\text{grid}} = \sqrt{N_{\text{objects}} \times 0.1}$, where N_{objects} is the number of objects inputted to the SOM algorithm. When selecting the prominent clusters, we only consider those that are consistently assigned to the same morphological class across all runs, thus guaranteeing that only unambiguous and robust classifications contribute to the final results, reducing the impact of any rigidity or bias inherent to the clustering algorithm.

(iii) We also consider only galaxies that are assigned the same label (disc or spheroid) in all runs, therefore mitigating the risk of preferentially clustering certain morphologies.

Lastly, to empirically assess the stability of our pipeline, we introduced a 5 per cent noise perturbation to the galaxy morphological metrics in each bin, prior to running the SOMBRERO algorithm. We find that the fraction of galaxies that changed their label across different runs is minimal (mean differences of 5 per cent for discs, 3 per cent for spheroids, and 4 per cent for irregulars), reinforcing that our methodology remains robust even when small systematic biases are included.

We show in Fig. 3 examples of galaxies from prominent clusters. We select 4 sets of 9 galaxies each, distributed as follows: top left and right – 18 galaxies (9 each) classified as spheroidal in the redshift ranges $0.2 \leq z < 0.4$ and $2.2 \leq z < 2.4$, respectively; bottom left and right – same as top panels, but showing systems classified as discs.

3.3 Unsupervised optimization

Two out of the four metrics have tunable parameters, which can affect their performance in separating disc and spheroidal galaxies. While M_{20} and G do not contain any tunable parameters, G_2 depends on module tolerance (m_{tol}) and phase tolerance (p_{tol}), and E depends on the number of bins (n_{bins}) (see Section 3.2 of K24). In order to avoid making assumptions about the underlying distributions, we adopt an unsupervised method to define m_{tol} , p_{tol} , and n_{bins} that best discriminate galaxies according to their morphological features.

For this optimization, we use the ‘knee value’ obtained from SOMBRERO clustering. The method in question utilizes the clustering algorithm, which naturally aims at discriminating galaxies with distinct morphological features. To optimize each parameter, m_{tol} , p_{tol} , and n_{bins} , we test a range of possible values for these parameters. We then run the optimization for each value from this list and select the one corresponding to the highest ‘knee value’ – defined as the steepest fall of the within-cluster sum of squares (WCSS) given by

$$\text{WCSS} = \sum_{i=1}^N \sum_{j=1}^d (x_{ij} - \mu_{c(i,j)})^2, \quad (4)$$

where

- (i) N : Total number of data points (individual galaxies).
- (ii) d : Number of dimensions (features) of the data (metrics).

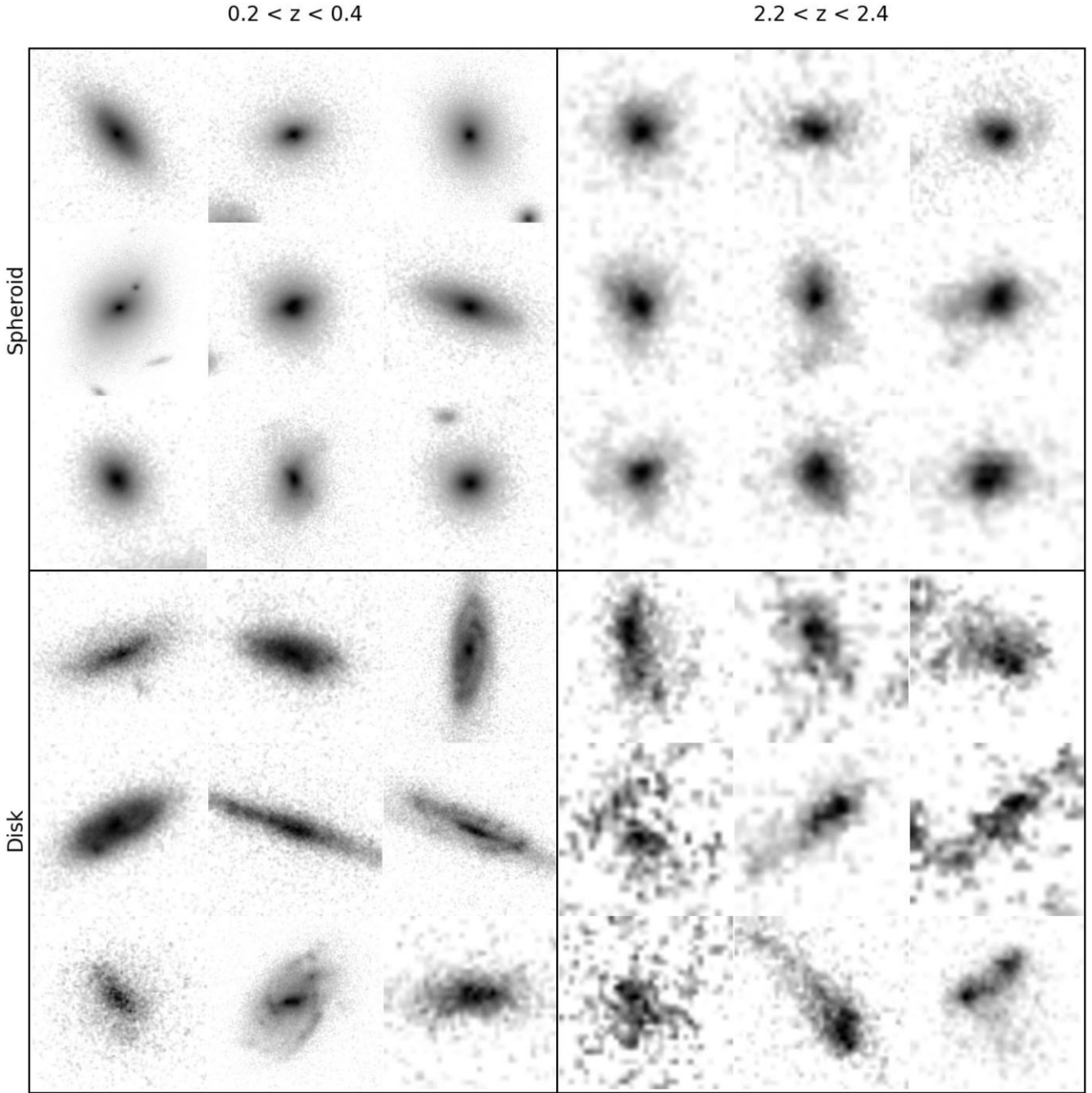


Figure 3. Examples of randomly selected galaxies from our data set basing on our model classification. This figure is split into four segments. Top left: 3×3 segment is showing nine galaxies classified as spheroidal at $0.2 < z < 0.4$ (first bin). Top right: 3×3 segment is showing nine galaxies classified as spheroidal at $2.2 < z < 2.4$ (last bin). Bottom left: 3×3 segment is showing nine galaxies classified as disc at $0.2 < z < 0.4$ (first bin). Bottom right: 3×3 segment is showing nine galaxies classified as disc at $2.2 < z < 2.4$ (last bin). We note how much different these higher redshift galaxies look like, especially in case of discs and how much more complex gets the task of the visual classification in higher redshifts.

- (iii) x_{ij} : Value of the j th feature of the i th data point.
- (iv) $\mu_{c(i)j}$: Value of the j th feature of the centre (mean) of the cluster $c(i)$ to which the i th data point belongs.
- (v) $c(i)$: Cluster assignment of the i th data point.

The same procedure simultaneously defines the optimal number of superclusters. This is possible because WCSS is calculated during the supercluster split, thus addressing two questions simultaneously: (1) the optimal parameter that demonstrates the lowest WCSS value

for a given supercluster number, and (2) the optimal number of superclusters. Interestingly, we consistently find two to be the optimal number of superclusters. We believe that this indicates that the current image resolution and metric sensitivity can split our data into only two morphological classes, supporting our claim that this methodology is suitable to separate galaxies into spheroidal and disc galaxies.

Fig. 4 exhibits how the WCSS function provides an optimal supercluster split for both metrics: G in Fig. 4(a) and G_2 in Fig.

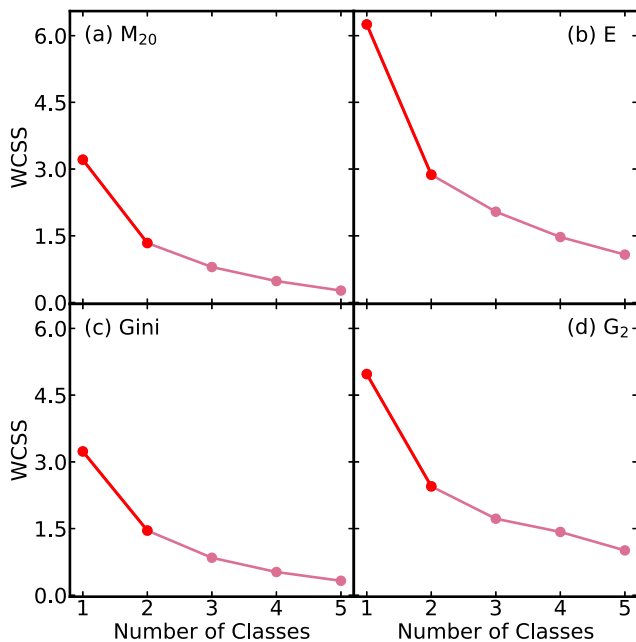


Figure 4. Comparison between metric capacity to split galaxies into multiple classes, at the same time indicating the optimal number of superclusters for each metric (signalized by bold segment). It is possible to note that slope of bold segment in each panel is a much steeper, indicating the higher change in WCSS, which in its turn points to a better supercluster split.

4(b). There is a steeper fall in the WCSS when the number of classes change from 1 to 2, in comparison to other transitions. Our rationale is based on the fact that if splitting the data into more superclusters significantly impacts the inter-cluster variance, indicated by a steeper decrease in WCSS, then the split is justified, and the current number of superclusters better describes the data. If the slope in the WCSS function is lower, it suggests that the split only marginally improves the variability, leading to the conclusion that there may not be sufficient resolution for this separation or the metrics used are not enough to find finer subclasses.

Additionally, the order in which we supply metrics to SOMBRERO can also influence results. In order to maximize the unsupervised clustering, we ensure that metrics with best discrimination between two classes will be prioritized (supplied first). At this stage, we have WCSS versus number of classes for each metric, quantifying its performance to split the data. We then order the metrics and supply to SOMBRERO. We find empirically that G_2 and E are the most impactful metrics according to WCSS, and they usually are supplied as first vector entries to SOMBRERO. G and M_{20} tend to score lower in terms of WCSS, and have less impact on clustering.

In summary, the unsupervised optimization consists of two steps: First, we run WSCC-based optimization for each metric individually, in order to obtain WCSS value for a given combination of parameters (n_{bins} for E , m_{tol} and p_{tol} for G_2); secondly, we run all the metrics together, but vary the parameters for the metric being optimized. By doing so, we guarantee that, in the second step, we evaluate not only the individual performance of each metric but also how it works in tandem with other metrics. We select the best parameters for G_2 and E based on this run. Likewise, we order the metrics based on WCSS resulting from the same run. In practical terms, this strategy results in two pipeline runs. The first run focuses on optimization, selecting the best values for n_{bins} for entropy, and m_{tol} and p_{tol} for G_2 . The

second run uses these optimized values to obtain metric values used as input for unsupervised clustering.

3.4 Deep learning and classification

Despite applying pre-processing steps to galaxy images for estimating non-parametric indices, we provide the original, unprocessed images to the DL algorithm. These are prepared by converting the initial $F814W$ filter cut-outs, originally in .fits format, to .png format using the TRILOGY package from Coe et al. (2012). Once labels are assigned by the SOMBRERO algorithm during the unsupervised stage, we move to the supervised approach with CNN models. Compared to unsupervised labelling, once a CNN model is trained, it can be applied much faster on vast amounts of data to provide labelling. However, it is important to remember that the CNN model is only as good as the data it was trained on.

In this study, we employ a popular form of transfer learning that involves using pre-trained DL models. These models, trained on large-scale data sets, are repurposed for specific tasks by replacing and fine-tuning the last few layers of the network. The underlying assumption of this approach is that earlier layers of deep networks learn general, reusable features (such as edges, patterns, and forms), while later layers become increasingly task-specific. Our methodology incorporates a transfer learning approach to achieve two main objectives: reducing the necessary data volume for effective training and expediting the training process by limiting the required epochs. We utilize pre-trained weights from the ImageNet data set (Deng et al. 2009). These weights, optimized to discern a plethora of features in the source domain, are aptly modified in our model to classify galaxies based on their morphological distinctions. With these pre-trained weights as a foundation, our model undergoes fine-tuning on our specific data set during training. This retains the high-level features learned from ImageNet while adapting to the specific intricacies of our galaxy classification task. As a result, our model achieves commendable accuracy using a significantly smaller training data set in a condensed time frame, underscoring the efficacy of the transfer learning paradigm.

For our pipeline, we use TENSORFLOW as our framework. This open-source ML framework, developed by the Brain Team at Google (Abadi et al. 2016), offers a comprehensive ecosystem that facilitates the development and implementation of a wide range of ML models and complex numerical computations. TENSORFLOW operates through data flow graphs where nodes represent mathematical operations and edges denote multidimensional data arrays – known as tensors – that flow between these nodes. For our primary model architecture, we employ XCEPTION, a DL model. ‘XCEPTION’, standing for ‘Extreme Inception’, is a modification of the Inception architecture. Its primary innovation lies in the use of depthwise separable convolutions, instead of the standard convolutions found in Inception. Comprising 36 convolutional layers, XCEPTION forms the feature extraction base of our model (Chollet 2017). This architecture allows us to leverage its advanced feature extraction capabilities, improving our model’s ability to classify galaxies accurately based on their morphological properties. The further steps in the pipeline are straightforward and unchanged compared to the original paper. We use the labelled data from SOMBRERO as the training set, splitting it into training and validation portions.

3.5 CNN confidence thresholding

According to our findings in K24, our labelled data set contains approximately 3–5 per cent uncertainty. In other words, some galax-

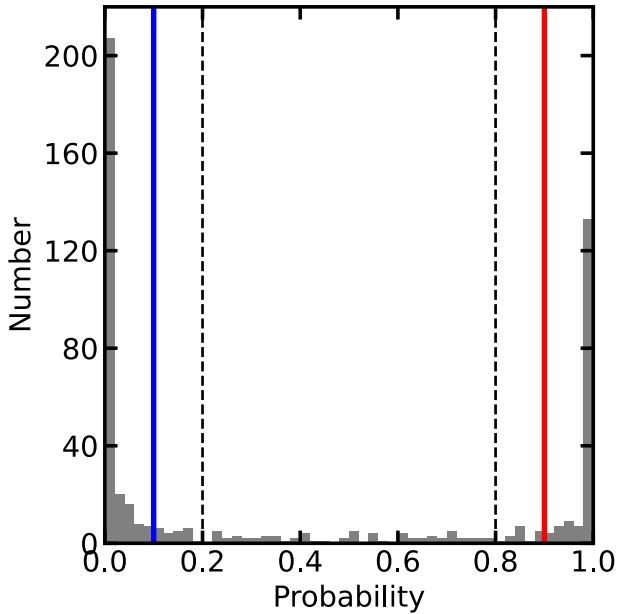


Figure 5. Confidence cuts applied to CNN prediction probabilities based on the return from the binary cross-entropy loss function, providing a single numeric value for each galaxy’s prediction. The figure displays the probability distribution from ensemble of 100 models of the first redshift bin. According to our labelling system, values close to 0 signify disc class (depicted in blue), and those close to 1 are identified as spheroids (depicted in red). At this stage, we introduce a class of irregular galaxies (depicted in two vertical dashed lines). These are galaxies that fall into the central region of the prediction probability distribution.

ies were mislabelled for various reasons and we address this issue employing an ensemble training technique, which involves training multiple models on identical data but with different random selections for the training and validation data sets. By using diverse subsets of the data for each individual model in the ensemble, we introduce a level of variation that can decrease the chance of overfitting and improve the robustness of the overall predictive performance. This ensemble strategy aims to leverage the strengths of multiple models, producing more stable and reliable predictions that are less influenced by uncertainty in the training data. Specifically, we train 100 models for each redshift bin. For each trained model, we select different galaxies to be part of training and validation data set. By this, we decrease the potential effect of misclassified galaxy having high impact on the final prediction, as all CNN probabilities will be averaged for all hundred models. This strategy not only mitigates the impact of uncertainty in the final classification but also provides us with information on the accuracy of the classifications. To further reduce the uncertainty in our training data set (one obtained from SOMBRERO clustering), we apply confidence thresholding to the final output of the DL classification. This methodology involves retaining data samples that exhibit a classification probability surpassing a predetermined high-confidence threshold. For instance, we use binary cross-entropy as our loss function. As the final result, this provides a classification probability, a single value ranged from 0 to 1. Values closer to 0 indicate a higher likelihood of the object being a disc galaxy, and values closer to 1 indicate a higher likelihood of the object being a spheroidal. Fig. 5 illustrates the distribution of the classification probabilities as they come from the CNN model. We note two peaks at the extremities, corresponding to the two classes, and a few intermediate values.

Therefore, we define galaxies with probability greater than 0.9 as spheroidal, and systems with probability smaller than 0.1 as disc galaxies, as shown by the red and blue lines in Fig. 5. We also define a sample of irregular galaxies, characterized by a probability between 0.2 and 0.8. Nevertheless, the results touching irregular galaxies should be treated with caution, as we here assume that the inability of our method in classifying a system as disc or spheroidal is an indication that the system presents irregular features compatible with irregular galaxies. Once we apply these probability thresholds to all bins to obtain labelled samples, a small portion of galaxies that do not fall into any classifiable range are referred to as unclassifiable and are removed from the data set for further studies.

4 CLASSIFYING GALAXIES IN A WIDE REDSHIFT RANGE

In this section, we explore how automated methods to classify galaxy morphology rely on redshift-related effects. Particularly, previous works use the same model to a wide redshift range. For instance, the redshift range 0.2–2.4 comprises a look-back time interval of 8.5 Gyr, which is considerably longer than the expected time-scale for galaxy star formation quenching and morphological transition (e.g. Sampaio et al. 2024). When considering the size and mass distribution, it is noticeable that early-type galaxies are, on average, smaller than late-type galaxies at all redshifts, with a faster average size evolution for early types and a moderate evolution for late types (van der Wel et al. 2014). Furthermore, galaxies become more concentrated at higher redshifts, suggesting that their formation may occur through mergers (Whitney et al. 2021). Therefore, it is expected that using a single model in a wide redshift range may directly affect the morphological classification. Furthermore, as shown in Fig. 6, the visual distinction between galaxies in the first and last redshift bins raises concerns about the reliability of visual classification and inspection.

4.1 Coarse versus fine-grain models

To avoid the pitfalls present on any DL modelling that use all galaxies over a wide redshift range as training data set, we apply the method described in Section 3 on each redshift bin shown in Fig. 2, separately. Ultimately, this means that we have a redshift-bin-dedicated model trained solely on the data labelled from SOMBRERO prominent clusters.

First, we test how a model trained only on low- z data performs when applied to higher redshift bins. To track accuracy when using different models to derive morphological classification, we use the F1 score – defined as the fraction of galaxies that do not change their class between comparisons. In Fig. 7(a), we show the F1 score when applying the model trained using only galaxies in the first redshift bin ($0.2 \leq z < 0.4$) to all other bins. The solid line alongside shaded regions represents the median and Q_σ scatter, calculated as

$$Q_\sigma = 0.743 \times (Q_{75 \text{ percent}} - Q_{25 \text{ percent}}), \quad (5)$$

where Q_x denotes the x th percentile, calculated using the ensemble of 100 models for each redshift bin. Throughout the paper, we calculate scatter in the desired quantities using equation (5). Exploring Fig. 7(a), we find a decreasing F1 score until $z \sim 1.2$, followed by a plateau at ~ 40 per cent. This is an important evidence that galaxy morphology is evolving with redshift, such that there are subtleties that need to be taken into account in a more fine-grain approach, instead of the coarse-grain usually found in recent literature.

Secondly, we test how a single model trained on all the galaxies in the sample, irrespective of redshift, compares to bin-dedicated

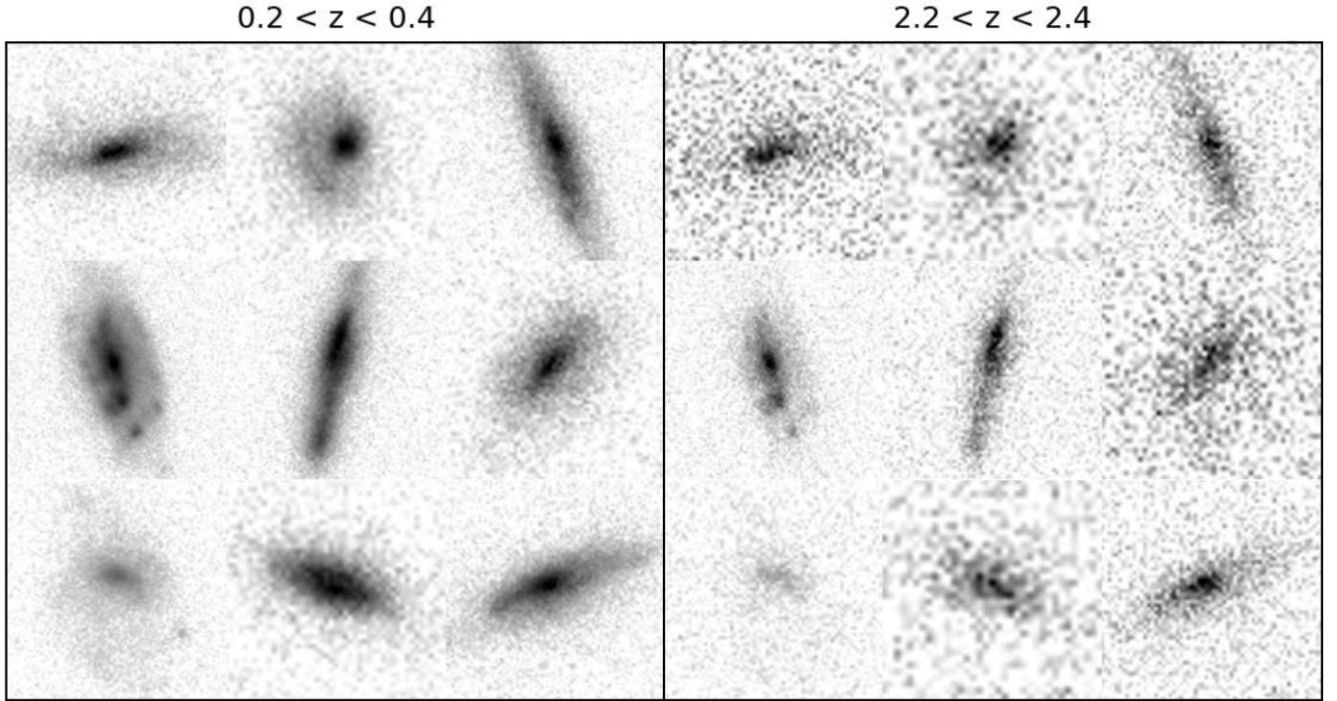


Figure 6. The figure presents several examples of the same galaxies that have been artificially redshifted. The left 3×3 mosaic displays the original selected disc galaxies, while the right shows these galaxies redshifted to a redshift of 2.3, corresponding to the last z bin. It is evident how significant the visual degradation effects are. This further confirms our concerns regarding visual classification and inspection; the majority of the features simply vanish or are no longer discernible.

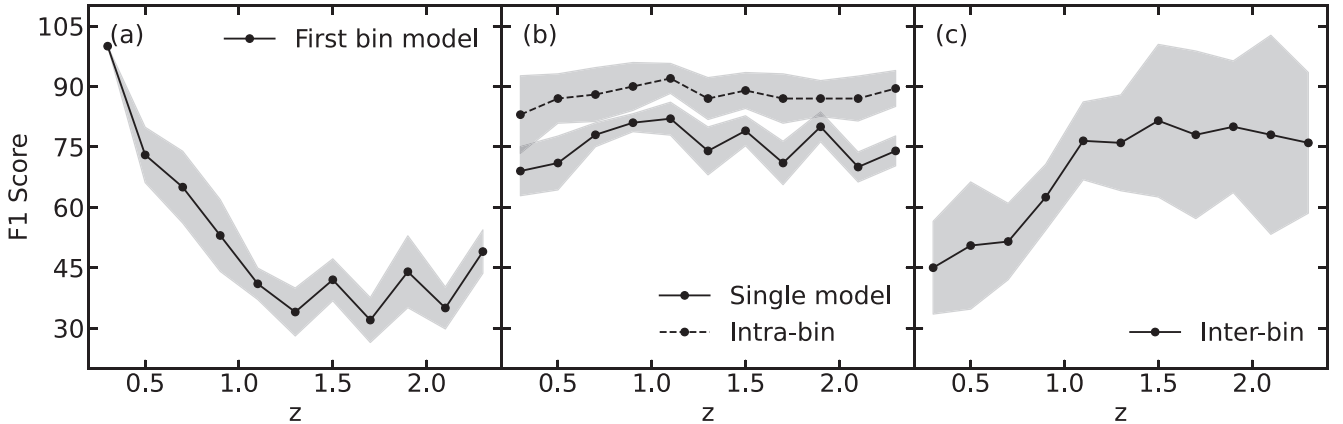


Figure 7. Showcase of model degradation across bins of z . (a) The ability of a model trained on the $0.2 < z < 0.4$ bin that classifies galaxies from all other bins. The figure shows expected results, where the performance falls rapidly as the galaxies change with higher z . (b) The performance, measured by F1 score, of the single model trained on prominent galaxies from all bins and compared with classification of in-bin models (solid black line) and performance (or inter-bin variability between ensemble of 100 models) of DL inside of specific bin. In other words, the dashed line shows the consistency of model training and application on different slices of data from same bin. We note only a 75 per cent match of classification between the single model and models dedicated to each bin across whole z range, while dashed line is closer to 90 per cent. Mean values are 75 and 87 per cent for single model and intra-bin, respectively. (c) Cross-matched application of models across whole z range. We note a stark decrease of match percentage in the lower redshifts, which agrees with findings in Conselice et al. (2011).

models. In Fig. 7(b), we show as solid line the F1 score when comparing a single model using galaxies with the one using bin-dedicated models. We find an F1 score of ~ 75 per cent, independent of the redshift, thus highlighting the importance of considering galaxy evolution and image degradation when assessing morphology. The change in the morphological classification of ~ 25 per cent of the galaxies indicates that applying a single model to a wide redshift

range can lead to biases due to specific morphological characteristics present at different redshifts. Nevertheless, it is noteworthy that, by attributing different models for each bins, we are also assuming that our model is capable of implicitly addressing degradation and evolution-related effects. Thus, we use the 100 ensembles in each redshift bin to investigate the variance in bin-dedicated models. The F1 score when comparing the intra-bin results is shown as dashed

line in Fig. 7(b). We find an average F1 score of ~ 90 per cent, which shows consistency in our morphological classifications. Comparison between solid and dashed lines shows that, even with variations in the bin-dedicated models, the F1 score is significantly higher ($\Delta F1$ score ~ 15 per cent) than adopting a single model to classify galaxies over the whole redshift domain.

Finally, in Fig. 7(c) we present a cross-match comparison between all the 11 bin-dedicated models. Namely, we classify galaxies in each redshift bin using the 11 bin-dedicated models. The comparison is done then between the classification attributed by the model dedicated to a given redshift bin, with the one assigned by the other 10 models. The cross-match comparison shows two relevant trends: (1) for the $z < 1$ bins, we find a decreasing F1 score, suggesting that these galaxies present morphological characteristics that possibly are not in the majority of galaxies in the $z > 1$ – for instance, we find an F1 score of 45 ± 11.5 per cent for our first redshift bin; (2) a different trend is observed for the $z \geq 1$ bins, where we find an almost constant F1 score of $\sim 75 \pm 17$ per cent – this trend indicates that the majority of models above this limit are considerably similar, despite a larger $1 - \sigma$ scatter. The last trend may follow from galaxies suffering a significant morphological variation around $z \sim 1$, in agreement with previous works (e.g. Conselice et al. 2011). Therefore, it is important that these changes are addressed correctly when defining a training set for a wide redshift range. Thus, we advocate a word of caution when dealing with automated algorithms for measuring galaxy morphology applied to a wide redshift range (e.g. Hausen & Robertson 2020; Robertson et al. 2023). Finally, we hereon present results using bin-dedicated models to provide morphological labels to galaxies in our sample, following the thresholds shown in Fig. 5.

4.2 Degradation and evolution of discs observed at high redshifts

When exploring higher redshift domains, numerous effects influence the observations. For instance, surface brightness dimming might obscure the fainter regions of a galaxy outskirts. The obscuration of discs may lead to biased classifications. Furthermore, since disc galaxies exhibit a clumpy light distribution, the entire disc is not necessarily obscured due to flux dimming. A region-dependent obscuration may lead to an overestimation of irregular galaxies' fraction. In addition, due to the Universe expansion and increasing distance, galaxies have a decreasing angular size with redshift, resulting in decreasing number of pixels available for any analysis. Both effects are significantly impactful in our work, as we extend our classification up to $z = 2.4$, and they can introduce additional complexities to the extraction of metrics and the performance of CNN models.

To address how degradation plus evolution affect our morphological analysis, we select all disc galaxies in the first redshift bin ($0.2 \leq z < 0.4$) and simulate how they would look at higher redshifts, while keeping the same instrumental set-up (exposure time, broad-band filters, and point-spread function), ranging from $z = 0.3$ to 2.3 in steps of 0.2. We use the Full and Efficient Redshifting of Ensembles of Nearby Galaxy Images (FERENGI; Barden, Jahnke & Häußler 2008) code to investigate the effect of surface brightness dimming and decreasing angular size. Briefly, the code simulates the observation of galaxies originally located at redshift z_0 to a desired redshift z' . Mathematically, the decreasing angular size is described by

$$\frac{a_{z'}}{a_{z_0}} = \frac{d_{z_0}/(1+z_0)^2}{d_{z'}/(1+z')^2}, \quad (6)$$

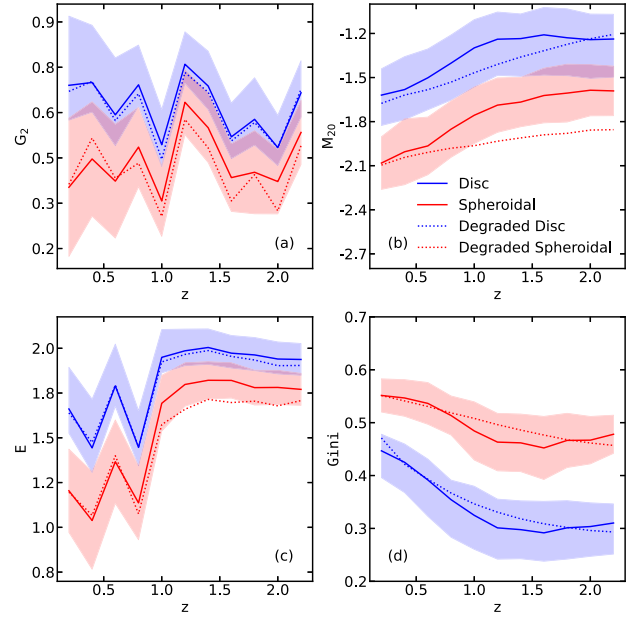


Figure 8. Metrics measured on degraded galaxies in comparison with the real disc and real spheroidal galaxies across the entire redshift range based on their metric values. Blue and red solid lines correspond to median value of main two classes. Blue and red dotted lines correspond to degraded set of galaxies of each class. In case of Gini and M_{20} , we note slight deviation, especially in the middle of the redshift range. In general, this behaviour reinforces the confidence in our degradation procedure.

where the left-hand side is the ratio between the angular size at z' and z_0 , and $d_{z'}$ and d_{z_0} are the luminosity distance at the same redshifts. Additionally, by imposing the same absolute magnitude in both redshifts, the dimming in flux follows

$$\frac{f_{z'}}{f_{z_0}} = \left(\frac{d_{z_0}}{d_{z'}} \right)^2, \quad (7)$$

where $f_{z'}$ and f_{z_0} are the flux observed at each redshift. Notably, the code also allows for a parametrization of the luminosity evolution of galaxies. This is particularly relevant to our work, as we touch redshifts consistent with the peak in cosmic star formation density, $z \sim 2-3$ (Madau & Dickinson 2014). We use an absolute magnitude correction, following the linear relation $M_{z'} = M_{z_0} - (1 \times z')$, where $M_{z'}$ and M_{z_0} are the absolute magnitude at z' and z_0 (Barden et al. 2008), respectively. We fix the evolution term as -1.0 simply as a way to show the impact of it. However, establishing a more accurate functional form for evolution is beyond the scope of this investigation, especially considering its uncertainty (Sandage 1961; Poggianti 1997; Contardo, Steinmetz & Fritze-von Alvensleben 1998; Abraham & van den Bergh 2001). For completeness, we performed the same exercise for spheroidal galaxies and present the degradation results alongside.

We calculate the MEGG metrics for the redshifted galaxies to ensure consistency with real galaxies from corresponding redshift bins. Fig. 8 illustrates the metrics derived from redshifted galaxies compared with those from real galaxies in corresponding bins. It is noteworthy that, while we do not note relevant metric variations with redshift, the simulated galaxies show consistent trends with real galaxies. More importantly, we note larger variations in M_{20} and G index, in comparison to G_2 and E. We thus compare the relevance

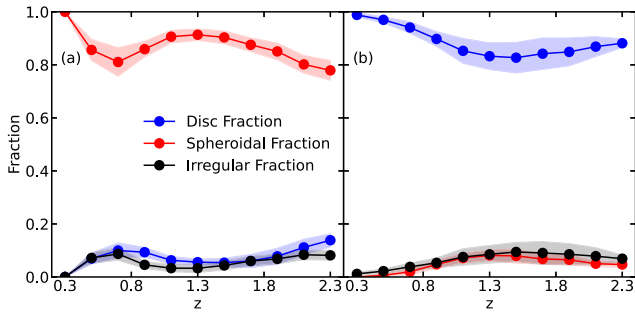


Figure 9. Degradation results for two main classes: spheroidal in panel (a) and disc in panel (b). Each panel contains fractions of disc, spheroidal, and irregular galaxies (from the simulated sample only) across our redshift range. It can be noted that both classes showcase a slight decrease at the beginning and then stabilize around 85 percent, while similar increase of irregular and spheroidal galaxies is noted. This indicates that, while some galaxies are misclassified as these classes due to redshift effects, this number hovers around 15 percent. Mean fraction for spheroidal degradation is 92 percent and for disc is 95 percent, which are expected in prism of comparison of overall method accuracy achieved in K24.

of each metric for the definition of prominent clusters, for which we find that, not by chance, the last two are the most relevant.

Using a set of artificially redshifted galaxies, we investigate how degradation and luminosity evolution effects may lead to misclassification in galaxy morphology. We take all galaxies in the redshift bin $0.2 < z < 0.4$ classified as spheroidal or discs and artificially degrade them to simulate observations at higher redshift bins from 0.4 to 2.4. We then classify the degraded galaxies using models dedicated to each specific redshift bin, constructed from real data – that is, we use models trained on data from Fig. 2. Fig. 9 presents the results for the two separate classes: spheroidal galaxies (in red, Fig. 9a) and discs (in blue, Fig. 9b). After performing inference with our dedicated models, we measure how many galaxies maintain their classes, defining the fraction displayed in the figure. Notably, both classes demonstrate similar behaviour, which can be largely attributed to the uncertainty of the method, resulting in mean fractions of 92 per cent for spheroidal and 95 per cent for discs. As presented in K24, we expect an accuracy of approximately 90 percent. The observed decrease of about 15 percent on average reinforces how spheroidal and disc galaxies can be misclassified simply due to degradation.

5 RESULTS AND DISCUSSION

While in the local Universe the morphology of galaxies can be easily classified by visual inspection – therefore defining a reliable training set, at higher redshifts flux dimming, observational limitations, and the presence of irregular features input a higher degree of subjectivity in visual inspection. Our work is focused on providing a reliable classification that does not depend on training sets built by visual classification. Next, we present our main results derived by applying the method described in Section 3 to CANDELS galaxies.

A particular advantage of analysing a wide redshift range is to probe how galaxies evolve over a large look-back time interval. In particular for morphology, the fraction of disc, spheroidal, and irregular galaxies have been the focus in the recent literature due to its implication to galaxy formation and evolution theory (Cassata et al. 2005; Tasca et al. 2009; Conselice et al. 2011; Mortlock et al. 2013; Talia et al. 2014; Kartaltepe et al. 2015; Lee et al. 2024). Therefore, we show in Fig. 10, respectively from top to bottom

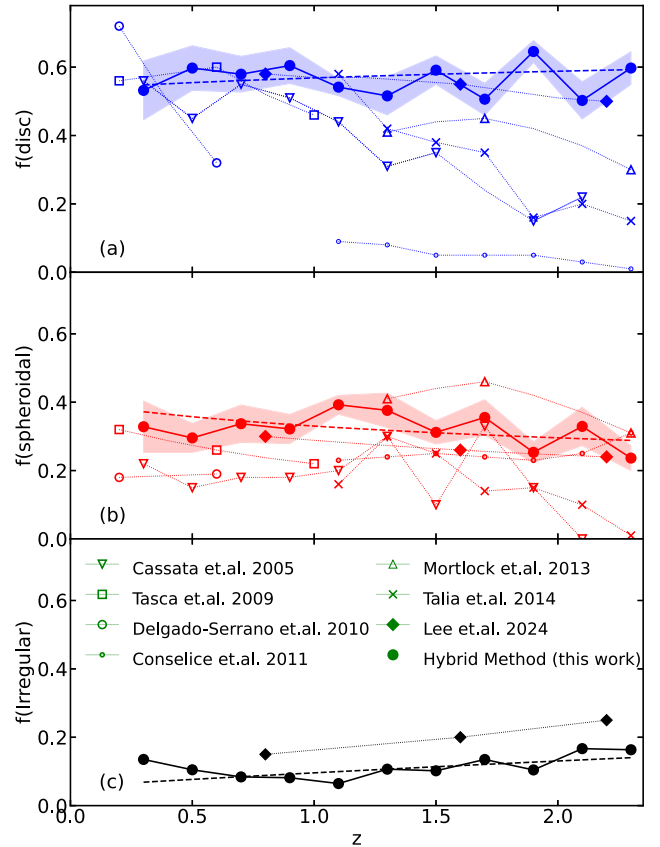


Figure 10. Comparison of the fractions obtained in this research with ones presented in the literature over the last decade. These fractions are plotted across the redshifts for disc, spheroidal, and irregular galaxies. It is notable that we tend to agree with most of the findings, for both main classes, reinforcing the robustness of the method. The dashed lines show the result of fitting the data to equation (8). To produce the error bars for our ratios, we used performance of each of our 100 ensemble models to calculate Q_σ and median. Tabular fractions can be found in Table A1. Summary of methodologies employed in different papers cited in this figure can be found in Table A3.

panels, the fraction of disc (blue), spheroidal (red), and irregular (black) galaxies according to our method (solid line), and according to previous works in the literature (dotted lines). Shaded areas around the solid line represent the scatter in our estimate calculated using 100 ensembles. Empty symbols represent works that make use of visual classification, whereas filled symbols show the results from works that adopt objective – i.e. do not depend on visual inspection – criteria to define morphology. In the case of irregular galaxies’ fraction, we only include the results from Lee et al. (2024), as the definition of irregular galaxies can be somewhat ambiguous and this is the only work in which the adopted definition is consistent with ours, i.e. follow from objective criteria.

Examining Fig. 10, we find that the fractions are highly dependent on the adopted method. In Table A3, we show a summary of the methods adopted in each work included in Fig. 10. Significant differences are observed in the $z > 1$ region, especially for discs, when comparing our results with those from visual inspection. This discrepancy could be even larger considering the effect shown in Fig. 9. Conversely, despite the use of different data and methodologies, our findings closely align with those in Lee et al. (2024), which also avoids visual inspection in morphological classification. We therefore suggest that visual inspection, whether directly or for

Table 1. Results for the fitting of equation (8) to the curves shown in Fig. 10. Column (1) shows the class considered. Columns (2) and (3) show the constant and exponent, alongside related uncertainties, for the fitting of equation (8) to the observed fractions.

Class	C_0	m_{class}
Disc	0.54 ± 0.05	0.08 ± 0.03
Spheroidal	0.39 ± 0.07	-0.26 ± 0.04
Irregular	0.06 ± 0.04	0.73 ± 0.05

creating training sets, should be avoided to prevent bias from the effects discussed in Section 4. Notably, while our classification relies on a single band (*F814W*), Lee et al. (2024) vary the observed band with redshift to always probe the same rest-frame wavelength for all galaxies, irrespective of redshift. Thus, the similarity in the results suggests minimal variations in the non-parametric indices across different wavelengths, even though the stellar population evolves significantly over the redshift range.

Comparison between disc, spheroidal, and irregular fractions over the interval $0.2 < z < 2.4$ shows an almost constant fraction for all three, irrespective of redshift. Discs comprise roughly ~ 60 per cent of galaxies, whereas spheroidal and irregular galaxies represent ~ 30 and ~ 10 per cent, respectively. We quantify variations in the fraction as a function of redshift using the following relation:

$$F(\text{class}) = C_0 \times (1 + z)^{m_{\text{class}}}, \quad (8)$$

where ‘class’ stands for discs, spheroids, or peculiars. We show in Table 1 the fitting results for the curves shown in Fig. 10.

Regarding disc fraction, our fitting results ($m_{\text{disc}} = 0.08$) reinforce an almost flat fraction as a function of redshift. Notably, this is in disagreement with most observations characterizing galaxies morphology beyond $z > 1$ (Cassata et al. 2005; Tasca et al. 2009; Delgado-Serrano et al. 2010; Mortlock et al. 2013; Talia et al. 2014), which suggests a decreasing fraction of discs towards higher redshifts. We suggest that this disagreement follows from limitations of visual inspection-based classification, as it gets increasingly harder to detect discs for increasing redshift.

With respect to irregular systems, we find a slight (~ 10 per cent) increase for the $z > 1$ region. Our fitting results suggest that the fraction of irregular galaxies increases with redshift, doubling its initial value after a $\Delta z \sim 1.57$. Despite, previous works also find an increasing fraction of irregular galaxies with redshift, our increase is considerably less steeper than previously reported. Altogether, we suggest that, when probing faint galaxies at high redshifts, discs may still have notable visible regions due to their clumpiness that, although not sufficient to characterize the galaxy as a disc, may induce an excess of irregular systems when classified by eye, simply because dimming makes the disc not visible but the knots in the disk characteristics of H II regions are identified by the eye.

Exploring results shown in Table 1, we find that, contrary to the trend observed for discs and irregulars, the fraction of spheroids decreases with redshift. Quantitatively, it decreases from ~ 36 per cent to ~ 20 per cent between $z = 0.2$ and $z = 2.4$, respectively. Therefore, we suggest that the formation of the majority of spheroidal galaxies may follow from evolution-related mechanisms, such as mergers (Bundy et al. 2009; Moody et al. 2014; Rodriguez-Gomez et al. 2017; Jackson et al. 2019), especially for massive systems, and active galactic nucleus/stellar feedback (Simpson et al. 2013; El-Badry et al. 2016; Zeng et al. 2024), particularly relevant for less massive systems, both previously pointed as triggers for morphological transition (e.g. Sampaio et al. 2022, 2024). Still, an

analysis of galaxy evolution as a function of redshift will be presented in a future work (Sampaio et al., in preparation).

Furthermore, galaxy evolution, including morphological evolution, depends on stellar mass. To assess the evolution of galaxy morphology as a function of stellar mass and redshift, we define three different redshift bins: $0.6 \leq z < 1.2$, $1.2 \leq z < 2.0$, and $2.0 \leq z < 2.4$, which are selected for consistency with Lee et al. (2024). Fig. 11 shows the fraction of spheroidal (red curve), disc (blue), and irregular (black) galaxies as a function of stellar mass for the three bins mentioned above. The shaded areas denote the $1 - \sigma$ scatter derived from the 100 ensembles (see Section 4). The dotted lines represent the results from Lee et al. (2024).

This comparison shows a striking similarity between our results and those of Lee et al. (2024), reinforcing the agreement when using objective methods to characterize morphology. Exploring the relation shown for irregular galaxies, we find an approximately constant fraction, irrespective of stellar mass and redshift. The only notable variation is in Fig. 11(c), which shows a slight decrease in the fraction of irregular galaxies for stellar masses above $10^{10.5} M_{\odot}$. Moreover, the most significant result comes from comparing disc and spheroidal galaxies. The fractions of discs and spheroids exhibit nearly opposite behaviours. In Fig. 11(a), the fraction of discs decreases by ~ 66 per cent over the entire stellar mass range, while the fraction of spheroids increases similarly by ~ 74 per cent. The same pattern is seen in Fig. 11(b), where the fraction of discs decreases by ~ 32 per cent, while the fraction of spheroids increases by ~ 34 per cent. In Fig. 11(c), the fraction of discs remains constant up to stellar masses of $10^{10.5} M_{\odot}$ and then decreases by ~ 7 per cent. The fraction of spheroids in this highest redshift bin shows an opposite trend compared to discs, constant up to stellar masses of $10^{10.5} M_{\odot}$ and then increasing by ~ 24 per cent.

It is important to emphasize that these trends are almost identical to those observed by Lee et al. (2024). The combined trends for discs and spheroids suggest that low-mass galaxies at low redshift are predominantly discs, in agreement with the downsizing scenario, assuming that most star formation occurs in disc galaxies. An important result from this investigation, also supported by Lee et al. (2024), is that morphological transformations over this redshift interval (~ 5 Gyr) depend on the stellar mass and redshift in a very complex manner. This may reflect the interplay of the several parameters defining the merger of two galaxies (e.g. Rodriguez-Gomez et al. 2015). Mergers are expected to play a fundamental role in the mass assembly of galaxies. Different works suggest that the merging of two discs is an efficient way to generate spheroidal galaxies (Toomre & Toomre 1972; Barnes & Hernquist 1992; Khochfar & Burkert 2003; Springel & Hernquist 2005; Hopkins et al. 2006) and the decreasing fraction of high-mass discs with redshift, alongside the increase of massive spheroidal galaxies, may indicate that galaxies evolve through mergers but the fractions we measure show a more complex picture.

6 SUMMARY

In this work, we classify 13 988 galaxies from the CANDELS fields into discs, spheroidal, and irregulars using a hybrid unsupervised-supervised method founded in non-parametric morphological metrics. To guarantee robustness, we apply $H \leq 24$ and $M_{\text{stellar}} \geq 10^9 M_{\odot}$ cuts to ensure reliability in the metrics’ estimation and avoid the inclusion of dwarf galaxies in our sample, respectively. In contrast to the commonly adopted CAS system, we use the EGG system – which was previously investigated and shows a better performance in discriminating spheroids and discs. We further include the M_{20}

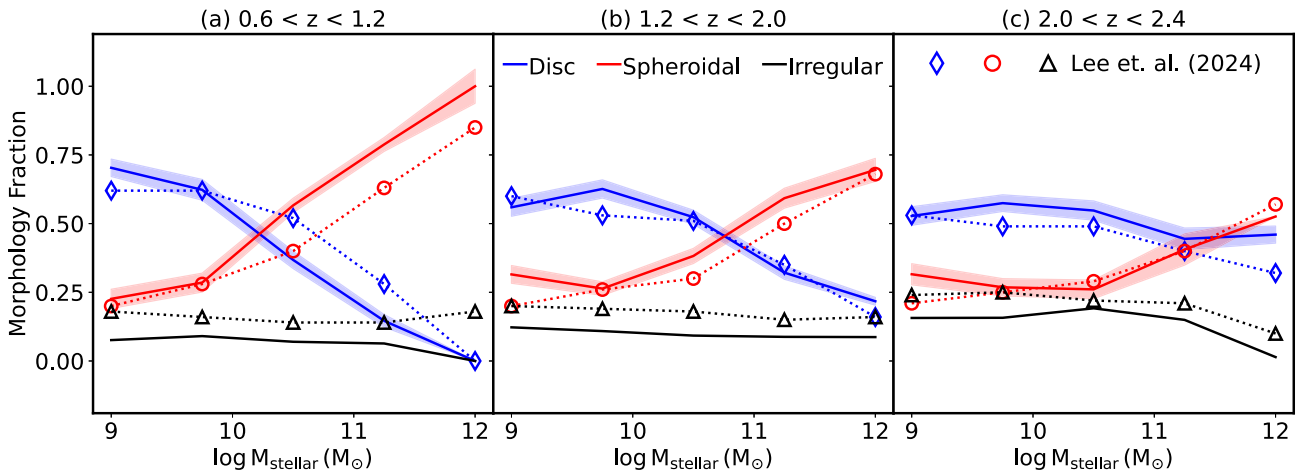


Figure 11. Companion figure to Fig. 10. Morphology fractions as a function of stellar mass in three segments of redshift are shown. Solid lines correspond to hybrid method results, while dotted lines with markers correspond to the results of Lee et al. (2024). It is noteworthy the compatibility of the results given that two researches used different data sets (*HST* and *JWST*) and different methods. The only point of intersection is that both methods are unsupervised-supervised and data driven. To produce the error bars for our ratios, we used performance of each of our 100 ensemble models to calculate Q_σ and median. Important note that errors in this figure are compatible with the error presented in Fig. 10. Tabular fractions can be found in Table A2.

metric, as it also shows good performance, therefore defining the MEGG system.

Differently from previous works, our method does not use visual inspection in any step. We first use the metrics as input to an SOM algorithm to define two prominent samples of disc and spheroidal galaxies. Then, this sample is used to train a CNN algorithm to provide final morphological classification. Furthermore, we provide an in-detail analysis of the limitations and biases that may occur in morphological classification when applying CNN models created over a wide range of redshift in comparison to bin-dedicated models. In this regard, one of our main conclusions is that bin-dedicated models show a better accuracy in comparison to a model built using data from the whole redshift range (Fig. 7.) We argue that this follows from the whole redshift range covering a look-back time much larger than the expected for galaxy evolution-related processes, thus increasing the inaccuracy of automated methods. Thus, we separate galaxies in redshift bins from 0.2 to 2.4, in steps of 0.2, with the step width selected to guarantee at least 500 objects in each bin.

Our results offer a new perspective on morphological evolution as a function of redshift and highlight the potential biases introduced by methods that rely on visual inspection. Here, we highlight the main findings of this work:

(i) By using the FERENGI code, we show that up to 17 percent of disc galaxies (Fig. 9) may be misclassified due to flux dimming, decreasing size, and luminosity evolution-related effects at redshifts $z > 1$. This should be taken into account when analysing the fraction of galaxies with different morphology, as it may introduce several biases in visual classification.

(ii) Contrary to studies suggesting a decreasing fraction of discs beyond $z > 1$, our morphological analysis results in disc and spheroidal fractions approximately constant irrespective of redshift, in excellent agreement with Lee et al. (2024). Quantitatively, the fraction of discs is significantly higher (~ 60 per cent) than the fraction of spheroidal galaxies (~ 30 per cent) over the redshift interval we study. Comparison between disc and spheroidal fractions suggests a ratio of approximately 1:2.

(iii) Comparison between fractions as a function of both redshift and stellar mass (Fig. 11) reveals a distinct trend for disc and spheroidal galaxies. We find a decreasing fraction of massive spheroidal galaxies with redshift, in comparison to an increasing fraction of massive disc systems. We suggest this complementary trend as an indication of the role of mergers in galaxy evolution. Namely, different works point out that mergers are an efficient way to generate spheroidal galaxies. However, the determined fractions display the intricacies of the merger history, meaning that this process is probably more complex than a simple transformation.

ACKNOWLEDGEMENTS

This work was financed in part by the Coordenação de Aperfeiçoamento de Pessoal de Nível Superior – Brazil (CAPES) – Finance Code 001. VMS and RRdC acknowledge the support from FAPESP through the grants 2020/15245-2 and 2020/16243-3, respectively. CC acknowledges support from the ERC Advanced Investigator Grant EPOCHS (788113).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author, IK, upon reasonable request.

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APPENDIX A: FRACTIONS OF THE DIFFERENT MORPHOLOGICAL TYPES AS PRESENTED IN FIGS 10 AND 11

Appendix provides fractions provided earlier in tabular format for readers' convenience.

Table A1. Table of fractions corresponding to Fig. 10. For each redshift bin (z), we provide the fractions of disc (D), spheroidal (S), and irregular (I) galaxies, along with their respective uncertainties, shown in the Q_{σ_D} , Q_{σ_S} , and Q_{σ_I} columns.

z	D	Q_{σ_D}	S	Q_{σ_S}	I	Q_{σ_I}
0.3	0.53	0.09	0.33	0.08	0.14	0.02
0.5	0.60	0.07	0.30	0.04	0.10	0.02
0.7	0.58	0.05	0.34	0.05	0.08	0.01
0.9	0.60	0.05	0.32	0.04	0.08	0.01
1.1	0.54	0.03	0.39	0.03	0.06	0.01
1.3	0.52	0.06	0.38	0.05	0.11	0.01
1.5	0.59	0.04	0.31	0.03	0.10	0.01
1.7	0.51	0.05	0.36	0.05	0.13	0.01
1.9	0.65	0.03	0.25	0.03	0.10	0.01
2.1	0.50	0.05	0.33	0.06	0.17	0.02
2.3	0.60	0.05	0.24	0.04	0.16	0.02

Table A2. Table of fractions corresponding to Fig. 11. For each redshift range (z) and corresponding stellar mass bin [$\log M_{\text{stellar}} (M_{\odot})$], we provide the fractions of disc (D), spheroidal (S), and irregular (I) galaxies, along with their respective uncertainties, shown in the Q_{σ_D} , Q_{σ_S} , and Q_{σ_I} columns.

z	$\log M_{\text{stellar}} (M_{\odot})$	D	Q_{σ_D}	S	Q_{σ_S}	I	Q_{σ_I}
0.6 < z < 1.2	9.00	0.70	0.03	0.23	0.03	0.08	0.01
0.6 < z < 1.2	9.75	0.62	0.04	0.29	0.03	0.09	0.01
0.6 < z < 1.2	10.50	0.37	0.03	0.57	0.03	0.07	0.01
0.6 < z < 1.2	11.25	0.15	0.02	0.79	0.03	0.06	0.01
0.6 < z < 1.2	12.00	0.00	0.00	1.00	0.06	0.00	0.06
1.2 < z < 2.0	9.00	0.56	0.03	0.32	0.03	0.12	0.01
1.2 < z < 2.0	9.75	0.63	0.03	0.26	0.02	0.11	0.01
1.2 < z < 2.0	10.50	0.52	0.03	0.38	0.03	0.09	0.01
1.2 < z < 2.0	11.25	0.32	0.02	0.59	0.04	0.09	0.02
1.2 < z < 2.0	12.00	0.22	0.01	0.70	0.04	0.09	0.06
2.0 < z < 2.4	9.00	0.53	0.03	0.32	0.04	0.16	0.02
2.0 < z < 2.4	9.75	0.57	0.03	0.27	0.03	0.16	0.02
2.0 < z < 2.4	10.50	0.55	0.03	0.26	0.03	0.19	0.02
2.0 < z < 2.4	11.25	0.44	0.04	0.41	0.06	0.15	0.04
2.0 < z < 2.4	12.00	0.46	0.03	0.53	0.00	0.01	0.00

Table A3. Table of methods used in papers quoted in Fig. 10. All the papers rely on visual inspection as crucial part of their classification.

Paper	Methodology	Components
Cassata et al. (2005)	Visual inspection; Sérsic index; CAS	(i) Visual class; (ii) parametric; (iii) non-parametric
Conselice et al. (2011)	Visual inspection; Sérsic index; CAS	(i) Visual class; (ii) non-parametric; (iii) parametric
Delgado-Serrano et al. (2010)	Visual inspection; bulge to total ratio	(i) Decision tree; (ii) visual class
Mortlock et al. (2013)	Visual inspection; CAS	(i) Visual class
Talia et al. (2014)	Visual inspection; CAS	(i) Visual class; (ii) non-parametric
Tasca et al. (2009)	Visual inspection; C + A + Gini	(i) Non-parametric; (ii) parametric; (iii) visual class

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